

Question1.js:

```
//Question1 - Online Store Sales Report

const orderAmounts = [1200, 450, 3000, 750, 1500, 250, 4200, 900];

//First order amount greater than 2000
const firstOrderGreaterThan2000 = orderAmounts.find(amount => amount > 2000);

//All orders greater than 1000
const ordersGreaterThan1000 = orderAmounts.filter(amount => amount > 1000);

//Total of all order amounts
const totalSales = orderAmounts.reduce((total, amount) => total + amount, 0);

//Display results
console.log("Original Order Amounts:", orderAmounts);
console.log("First Order Greater Than 2000:", firstOrderGreaterThan2000);
console.log("Orders Greater Than 1000:", ordersGreaterThan1000);
console.log("Total Sales:", totalSales);
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports
> Original Order Amounts: (8) [1200, 450, 3000, 750, 1500, 250, 4200, 900]
  First Order Greater Than 2000: 3000
> Orders Greater Than 1000: (4) [1200, 3000, 1500, 4200]
  Total Sales: 12250
```

Question2.js:

```
//Q2: Student Performance Analyzer

const marks = [45, 78, 92, 61, 88, 54, 73, 95];

//Increase every mark by 5
const increasedMarks = marks.map(mark => mark + 5);

//students whose marks are 70 or above
const marks70OrAbove = marks.filter(mark => mark >= 70);

//First mark greater than 90
const firstMarkGreaterThan90 = marks.find(mark => mark > 90);

//Total of all original marks
const totalMarks = marks.reduce((total, mark) => total + mark, 0);
```

```
//Display results
console.log("Original Marks:", marks);
console.log("Marks After Increasing by 5:", increasedMarks);
console.log("Marks 70 or Above:", marks70OrAbove);
console.log("First Mark Greater Than 90:", firstMarkGreaterThan90);
console.log("Total Original Marks:", totalMarks);
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports

> Original Marks: (8) [45, 78, 92, 61, 88, 54, 73, 95]
> Marks After Increasing by 5: (8) [50, 83, 97, 66, 93, 59, 78, 100]
> Marks 70 or Above: (5) [78, 92, 88, 73, 95]
  First Mark Greater Than 90: 92
  Total Original Marks: 586
```

Question3.js:

```
//Q3: Employee Data from an API

const employees = [
  { name: "Ali", department: "IT", salary: 80000 },
  { name: "Sara", department: "HR", salary: 70000 },
  { name: "Ahmed", department: "IT", salary: 95000 },
  { name: "Ayesha", department: "Finance", salary: 85000 }
];

// Employees belonging to IT department
const itEmployees = employees.filter(employee => employee.department === "IT");

// First employee with salary greater than 90000
const highSalaryEmployee = employees.find(employee => employee.salary > 90000);

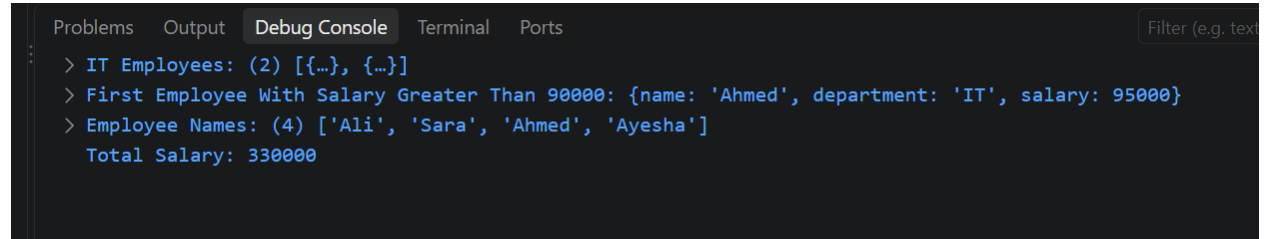
// List containing only employee names
const employeeNames = employees.map(employee => employee.name);

// Total salary
const totalSalary = employees.reduce(
  (total, employee) => total + employee.salary,
  0
);

// Display results
console.log("IT Employees:", itEmployees);
```

```
console.log("First Employee With Salary Greater Than 90000:", highSalaryEmployee);
console.log("Employee Names:", employeeNames);
console.log("Total Salary:", totalSalary);
```

Output:



```
Problems Output Debug Console Terminal Ports Filter (e.g. text)
> IT Employees: (2) [{...}, {...}]
> First Employee With Salary Greater Than 90000: {name: 'Ahmed', department: 'IT', salary: 95000}
> Employee Names: (4) ['Ali', 'Sara', 'Ahmed', 'Ayesha']
    Total Salary: 330000
```

Question4.js:

```
// Question 4 - Store Inventory Using JavaScript Map

// Create inventory Map
const inventory = new Map([
  ["apples", 500],
  ["bananas", 300],
  ["oranges", 200]
]);

console.log("Initial Inventory:", inventory);

// Add a new product
inventory.set("mangoes", 150);

// Update quantity of an existing product
inventory.set("apples", 600);

// Retrieve quantity of a specific product
console.log("Quantity of Apples:", inventory.get("apples"));

// Check whether a product exists
console.log("Does bananas exist?", inventory.has("bananas"));

// Remove one product
inventory.delete("oranges");

// Display current number of products
console.log("Number of Product Entries:", inventory.size);

// Display all product names
console.log("Product Names:");
```

```

inventory.forEach((quantity, product) => {
    console.log(product);
});

// Display all quantities
console.log("Product Quantities:");

inventory.forEach((quantity) => {
    console.log(quantity);
});

// Display product and quantity together
console.log("Current Inventory:");

inventory.forEach((quantity, product) => {
    console.log(product + ": " + quantity);
});

// Empty the inventory
inventory.clear();

// Display final size
console.log("Final Inventory Size:", inventory.size);

```

Output:

```

> Initial Inventory: Map(3) {size: 3, apples => 500, bananas => 300, oranges => 200}
Quantity of Apples: 600
Does bananas exist? true
Number of Product Entries: 3
Product Names:
apples
bananas
mangoes
Product Quantities:
600
300
150
Current Inventory:
apples: 600
bananas: 300
mangoes: 150
Final Inventory Size: 0

```

Question5.js:

```
// Question 5 - Student Result Object

// First student object
const student1 = {
  name: "Fatima",
  class: "BSCS",
  chemistry: 85,
  mathematics: 90,
  Urdu: 80,
  English: 75,

  calculateResult: function () {
    const total =
      this.chemistry +
      this.mathematics +
      this.Urdu +
      this.English;

    const percentage = (total / 400) * 100;

    return percentage;
  }
};

// Display name and class using dot notation
console.log("Student Name:", student1.name);
console.log("Student Class:", student1.class);

// Access property using bracket notation
console.log("Chemistry Marks:", student1["chemistry"]);

// Call the method
const percentage1 = student1.calculateResult();

console.log(
  student1.name + "'s Percentage:",
  percentage1 + "%"
);

// Second student object
const student2 = {
  name: "Ayesha",
  class: "BSCS",
  chemistry: 90,
```

```

    mathematics: 85,
    Urdu: 88,
    English: 92,

    calculateResult: function () {
        const total =
            this.chemistry +
            this.mathematics +
            this.Urdu +
            this.English;

        const percentage = (total / 400) * 100;

        return percentage;
    }
};

// Call method for second student
const percentage2 = student2.calculateResult();

console.log(
    student2.name + "'s Percentage:",
    percentage2 + "%"
);

// Remove one property from first student
delete student1.Urdu;

console.log("Final Student 1 Object:", student1);

```

Output:

```

Problems  Output  Debug Console  Terminal  Ports  Filter (e.g. text, !exclude, ...
Student Name: Fatima
Student Class: BSCS
Chemistry Marks: 85
Fatima's Percentage: 82.5%
Ayesha's Percentage: 88.75%
> Final Student 1 Object: {name: 'Fatima', class: 'BSCS', chemistry: 85, mathematics: 90, English: 75, ...}

```